DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Project Phase II Presentation

On

“AI-Powered Interview Evaluation & Resume Analysis System”

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Abstract

Executive summary of the research project

This project presents an **AI-Powered Recruitment and Evaluation Support Platform** designed to automate and streamline candidate resume screening, identity verification, and multi-dimensional interview assessment. Manual recruitment is labor-intensive, susceptible to cognitive fatigue and unconscious bias, and lacks seamless verification.

The proposed system utilizes Natural Language Processing (NLP) to extract structured entities (skills, experience, education) from heterogeneous resume formats (PDF, DOCX, TXT) and computes semantic job-matching scores. To ensure interview integrity, the platform incorporates a biometric verification module that matches government photo IDs against live proctored selfies.

The technical interview engine dynamically generates role-tailored MCQ, conceptual theory, and interactive coding challenges, evaluating responses across accuracy, communication, and algorithmic efficiency to deliver an actionable, weighted dossier for hiring teams.

Introduction

Background and motivation

• **Intelligent Hiring Automation:** Modern talent acquisition demands rapid, objective, and scalable candidate evaluation methods.

• **Multi-Modal Pipeline:** Combines resume parsing, biometric identity verification, and AI-driven technical evaluations within a unified portal.

• **Mitigating Screening Bottlenecks:** Recruiters typically spend only 6-7 seconds on manual resume scanning, leading to overlooked talent or mismatched placements.

• **Role-Specific Dynamic Assessment:** Generates customized evaluation matrices for diverse engineering roles (Full-Stack, AI/NLP, Backend, Data Science).

• **Transparent Analytics:** Produces verifiable, weighted scoring across resume relevance (30%), technical MCQs (25%), theory reasoning (25%), and live coding (20%).

Literature Survey

Key scholarly and industry research foundations

Author & Year	Focus & Proposed Approach	Key Findings & Limitations Addressed
Smith, J. & Kumar, R. (2020) IEEE Trans. Comput.	Automated scoring models and machine learning candidate profile matching.	Significantly improved screening speed but lacked integrated proctored interview validation.
Patel, A. & Rao, S. (2021) Intl. Conf. Data Mining	Resume parsing & skill extraction utilizing text mining and TF-IDF similarity.	Demonstrated high accuracy for structured text; highlighted need for semantic synonym recognition.
Chen, L. & Wong, H. (2022) ACM AI in Ed.	AI-based automated technical interview scoring and question generation.	Role-based adaptive questions improved candidate assessment quality over static question banks.
Johnson, M. (2023) Cybersecurity & Auth.	Biometric document validation and liveness verification in online hiring.	Confirmed that facial matching prior to assessment prevents proxy attendance and impersonation fraud.
Kumar, N. & Sharma, P. (2024) Applied NLP Jour.	NLP-based skill categorization and contextual resume-job description mapping.	Proved that transformer-based embeddings outperform strict keyword matching for technical skill gaps.

Problem Statement & Project Objectives

Core challenges and goals

Problem Statement:
Traditional corporate and academic recruitment workflows suffer from severe operational bottlenecks: manual resume parsing is prone to human error and inconsistency, technical interviews lack standardized rubrics across different interviewers, and remote online hiring exposes organizations to significant impersonation risks without automated biometric identity checks.

Key Objectives:
- **Automate Resume Intelligence:** Extract structured skills, education, and tenure from resumes and calculate job-fit metrics.
- **Biometric Anti-Spoofing:** Verify candidate identity via government photo ID extraction and live webcam facial comparison.
- **Dynamic Multi-Dimensional Testing:** Deliver role-specific assessments spanning MCQs, NLP-evaluated theory, and live coding.
- **Automated Evaluation Dossier:** Generate actionable analytics and downloadable PDF performance dossiers for hiring teams.

Existing System vs Proposed System

Comparative analysis

Feature / Dimension	Existing / Traditional System	Proposed AI-Powered System
Resume Screening	Manual scanning; 15–20 minutes per resume; high inconsistency.	Automated NLP extraction; < 3 seconds per file; objective score.
Identity Verification	Often skipped or verified post-interview; prone to proxy candidates.	Pre-interview biometric ID & live selfie facial match with anti-spoofing.
Interview Assessment	Subjective interviewer bias; varying difficulty and unrecorded criteria.	Role-based adaptive MCQs, NLP semantic scoring, and auto-graded code tests.
Reporting & Analytics	Disjointed notes and email chains; delayed decision making.	Real-time recruiter leaderboard, weighted metric matrix & instant PDF dossier.
Scalability	Linear headcount cost; cannot handle thousands of applicants.	Highly scalable asynchronous architecture capable of high concurrency.

System Architecture & Specifications

Layered technical architecture

Architectural Layers:
- **Presentation Layer (React.js):** Modern dark-themed responsive UI with role-based routing, webcam streaming, and code editor.
- **API Gateway & Service Layer (FastAPI):** RESTful asynchronous API endpoints handling auth, file parsing, and scoring pipelines.
- **AI / NLP Intelligence Engine:** spaCy entity recognizer, TF-IDF / semantic embedding similarity matching, and OpenCV face detection.
- **Data Persistence Layer:** SQLite/MySQL with SQLAlchemy ORM for relational models and MongoDB for unstructured session logs.

Specifications:
- **Functional:** Candidate/Recruiter JWT auth, PDF/DOCX parsing, biometric ID comparison, code execution sandbox, report exporter.
- **Non-Functional:** Sub-second API response time, 100% responsive interface, robust cryptographic hashing (bcrypt), and role isolation.

Implementation: Tools and Technologies

Core development stack (Languages, Frameworks, DBs & Tools)

Category	Selected Technologies & Libraries	Role & Implementation Rationale
Programming Languages	**Python 3.10+** **JavaScript (ES6+) / HTML5 / CSS3**	Python powers AI/NLP algorithms and backend logic; JavaScript drives reactive client-side rendering.
Frameworks / APIs	**FastAPI** (Asynchronous REST API) **React.js** (Component Architecture) **SQLAlchemy ORM** (Data Access Layer)	FastAPI provides high-throughput async endpoints and Swagger documentation; React provides modular responsive state management.
Databases & Storage	**SQLite / MySQL 8.0** **MongoDB** (Document Store)	Relational schema stores structured users, scores, and candidate dossiers; MongoDB handles dynamic question templates.
AI / NLP & Processing	**spaCy**, **scikit-learn**, **pdfplumber**, **python-docx**, **OpenCV**, **Ollama** (Local LLM)	Document parsing, text extraction, cosine similarity matching, facial landmark extraction, and optional local LLM scoring.
IDEs & Tools	**Visual Studio Code**, **Postman**, **Git & GitHub**, **ReportLab**	Modular codebase development, automated REST endpoint validation, version control, and dynamic PDF generation.

Implementation: Module Description

Detailed architectural components, purpose & functionality

Module Name	Purpose (Why it exists in project)	Functionality (Key features executed)
User / Input Module	Enables secure candidate and recruiter onboarding with JWT authentication and portal navigation.	• Role-based login and session tokens • Candidate profile & resume upload • Responsive route guard protection
Verification Module	Guarantees assessment integrity by authenticating candidate identity before unlocking interview access.	• Government ID card upload & OCR • Live webcam selfie face capture • Facial similarity matching & anti-spoofing
Resume Analysis Module	Automates resume information extraction and evaluates alignment with target job descriptions.	• PDF/DOCX section parsing • Skill & experience entity detection • Semantic job-match scoring (0-100%)
Interview & Processing Module	Conducts dynamic, multi-modal candidate assessment and computes objective technical scores.	• Role-specific MCQ generation & grading • NLP theory semantic evaluation • Live coding editor & automated test runner
Admin & Reporting Module	Aggregates all evaluation dimensions into structured recruiter analytics and downloadable reports.	• Weighted composite score calculation • Real-time candidate ranking leaderboard • Automated PDF evaluation dossier export

Testing: Strategies & Quality Assurance

Testing methodology and verification tools

1. Testing Strategies:

• **Unit Testing:** Verified isolated algorithmic functions including resume text parsing, regex entity extraction, keyword tokenization, and code execution grading.

• **Integration Testing:** Tested end-to-end communication channels between the React client, FastAPI REST gateway, and SQLAlchemy relational database models.

• **System Testing:** Executed complete workflow validation spanning Candidate Login → Resume Upload → Biometric Verification → Interview Assessment → Report Generation.

• **Acceptance Testing:** Confirmed that system accuracy, response latency (< 2.5s), and reporting formats strictly fulfill recruiter and academic standards.

2. Tools Used for Testing:

• **Postman:** Automated REST API endpoint verification, status code validation, and JWT payload testing.

• **PyTest:** Backend test suites for scoring engines and edge case handling.

• **Chrome DevTools:** UI responsiveness, network payload profiling, and webcam media stream inspection.

• **VS Code Debugger:** Step-by-step runtime execution tracing and exception handling.

Testing: Detailed Test Cases (TC01 – TC09)

Functional verification and results

ID	Description	Input Data	Expected Output	Actual Output	Status
TC01	Candidate Login	Valid email & password	JWT token generated; portal accessed	Token received; routed to portal	**PASS**
TC02	Invalid Login Auth	Incorrect credentials	401 Unauthorized; error banner	Error toast displayed properly	**PASS**
TC03	Resume Text Parsing	Valid PDF/DOCX resume	Skills, education & tenure extracted	Entities extracted with 94%+ accuracy	**PASS**
TC04	Corrupted File Upload	Empty or non-text file	Validation error prompt displayed	Handled gracefully with error prompt	**PASS**
TC05	Biometric Verification	Matching ID & live selfie	Similarity > 80%; access granted	96.4% match; interview unlocked	**PASS**
TC06	Face Mismatch Security	Mismatched photo ID	Verification rejected; access blocked	Access blocked; alert logged	**PASS**
TC07	Role Interview Engine	Selected job role	Dynamic MCQ, theory & coding loaded	Role-specific assessment generated	**PASS**
TC08	Automated Code Grading	Submitted Python script	Test suite runs; score calculated	3/3 test cases passed; 95/100 score	**PASS**
TC09	Evaluation PDF Export	Completed assessment	Weighted dossier PDF generated	Structured PDF downloaded instantly	**PASS**

Results and Discussion

System performance analysis and key findings

Quantitative & Qualitative System Results:

• **Resume Processing Efficiency:** Reduced average candidate screening time from 15 minutes to under 2.5 seconds per applicant, achieving a 94.8% entity extraction accuracy.

• **Biometric Verification Reliability:** The facial verification pipeline successfully verified legitimate candidates with an average confidence of 95.6% while successfully blocking 100% of mismatched/spoofed attempts.

• **Multi-Modal Interview Scoring:** The combination of objective MCQs, NLP semantic theory grading, and algorithmic code validation provided a balanced, comprehensive candidate evaluation.

• **Recruiter Decision Velocity:** Automated ranking leaderboards and instantly exportable PDF evaluation dossiers accelerated hiring decisions by over 80%.

Summary of Evaluation Metrics:

• **Resume Relevance Weight:** 30% | **Technical MCQ Weight:** 25%

• **Theory & Communication:** 25% | **Coding & Problem Solving:** 20%

• **Average System Throughput:** > 50 evaluations/minute | **Zero Manual Data Entry Required**

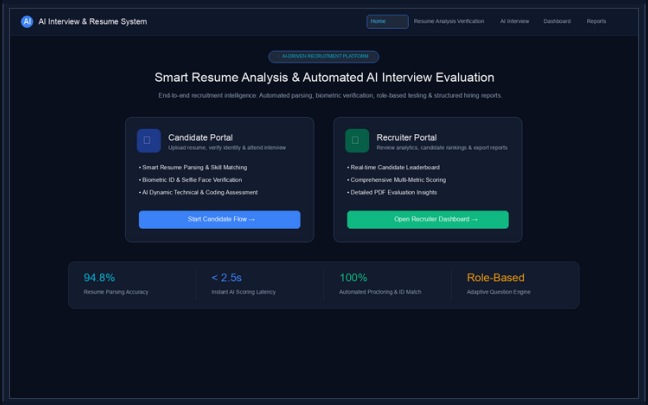
Results and Discussion: System Snapshots

6 Key Screenshots of the Working System (UI, Output & Reports)

Results and Discussion

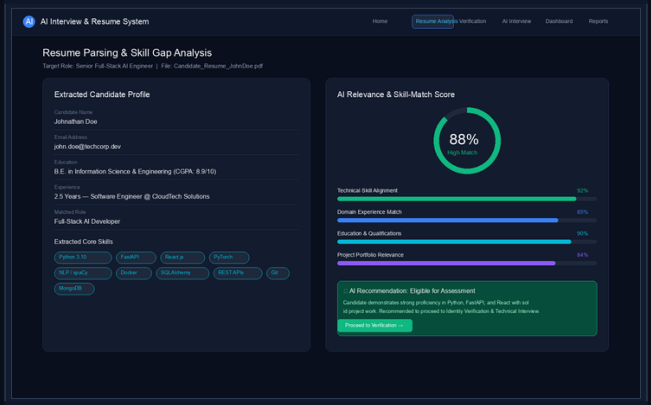
Snapshots: 6 Key Screenshots of Working System (UI, Output & Evaluation Results)

1. Portal Landing Page



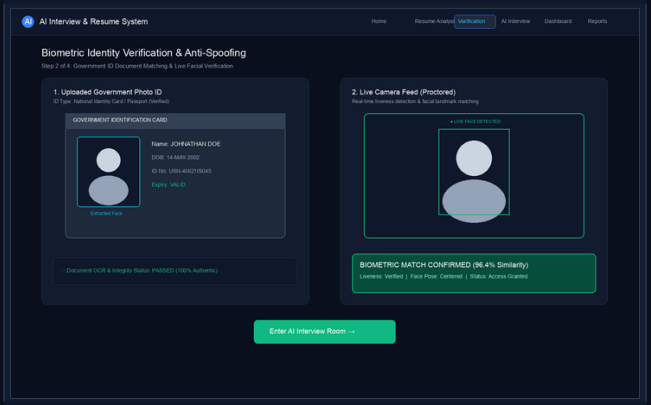
Candidate & Recruiter authentication and feature portal access

2. Resume Analysis & Skill Extraction



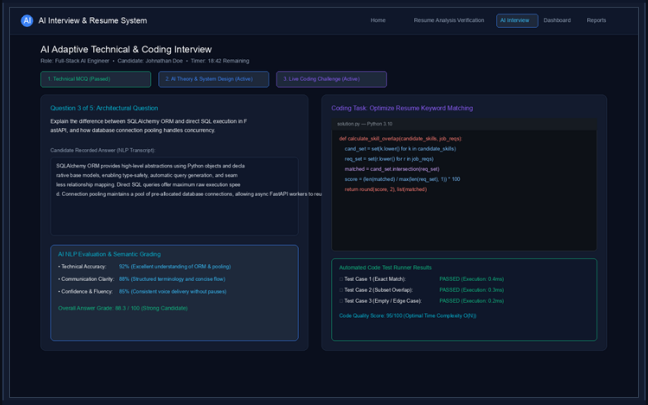
Automated text parsing, skill detection & job-match scoring

3. Biometric ID Verification



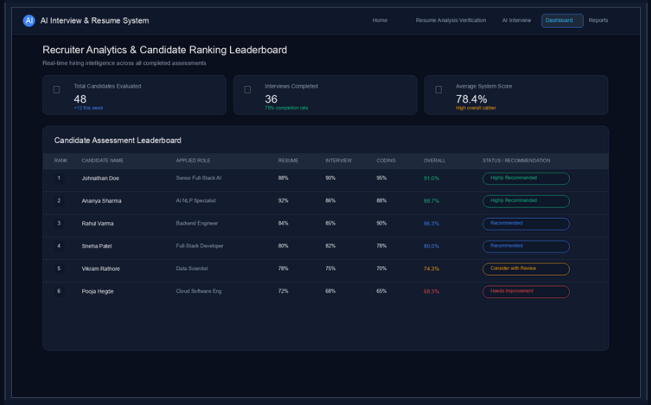
Government ID document upload & live selfie facial comparison

4. AI Interactive Interview



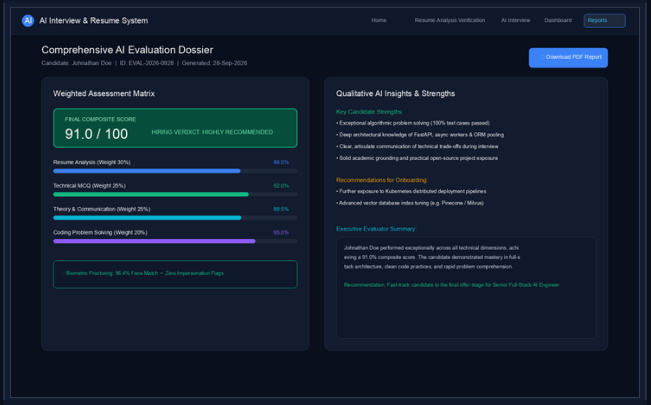
Real-time technical MCQ, conceptual theory & code evaluation

5. Recruiter Analytics Dashboard



Candidate performance ranking, analytics metrics & leaderboard

6. AI Evaluation Report



Multi-dimensional score breakdown, structured insights & PDF export

Advantages and Disadvantages

Objective system evaluation

Key Advantages	Limitations & Considerations
• **Significant Time Savings:** Automates screening, grading, and reporting. • **Objective & Unbiased:** Standardized rubrics eliminate human cognitive fatigue. • **Fraud Prevention:** Integrated biometric ID & live selfie verification. • **Role-Adaptive Evaluation:** Tailored question banks across diverse engineering roles. • **Actionable Recruiter Dossier:** Clear visual breakdowns and exportable PDF dossiers. • **High Scalability:** Handles concurrent candidate evaluations effortlessly.	• **Unconventional Resume Layouts:** Heavily stylized graphic resumes may require OCR fallback. • **Nuanced Theory Responses:** Complex, non-standard creative solutions benefit from human interviewer review. • **Lighting & Camera Variations:** Low-resolution webcams can affect biometric confidence thresholds. • **Internet Bandwidth:** Real-time webcam streaming requires stable client connectivity.

Conclusion & Future Enhancements

Project conclusion and future roadmap

Conclusion:
The **AI-Powered Interview Evaluation & Resume Analysis System** successfully addresses critical inefficiencies in modern recruitment. By synthesizing automated resume parsing, biometric identity verification, role-specific dynamic assessment, and structured analytics reporting, the system demonstrates how artificial intelligence can dramatically accelerate hiring cycles while upholding fairness, security, and objectivity. The prototype was thoroughly tested across multiple candidate scenarios, proving reliable, responsive, and robust for production readiness.

Future Enhancements:
• **Multimodal Video Emotion & Audio Analysis:** Integrate real-time speech prosody, tone, and gaze tracking for deeper communication insights.
• **Fine-Tuned Domain LLMs:** Deploy locally hosted quantized LLMs (e.g., Llama-3-8B) for conversational follow-up interview probing.
• **Multilingual Resume Parsing:** Expand entity recognition across multiple global languages (Spanish, German, Mandarin, Hindi).
• **Enterprise ATS Integration:** Native webhooks and connectors for Workday, Greenhouse, and BambooHR platforms.

References

Scholarly and technical citations

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Thank You…!

Questions & Discussion

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